

Who is buying MSOs in India

Embedded designers are the key users of MSOs. These applications span across several segments, from communications, semiconductors, computing, aerospace defence, industrial electronics, consumer electronics and automotive industries.

Educational institutes also find MSOs of great value as these help students visualise the intricacies of digital designs. Several reference boards and instruction aids, commonly used in college labs, deploy serial buses on these. Serial bus triggering and decoding helps students learn and understand the internals of embedded designs.

—Karl Chen, product manager, Rigol Technologies Inc.



Creative use case of a mixed-signal scope for a specific application

Matthew Navarro is a deployed US Navy sailor aboard USS Gridley, a guided-missile destroyer. But being an electronics wizard aboard a ship has inherent challenges. Navarro said, "We regularly have to run around the ship and perform repairs on installed equipment. The only oscilloscope we have for functional checking and measurement is a 20.4kg (45-pound) benchtop unit and we are out of luck if there is no electrical socket."

He was able to replace what he accurately termed a 'behemoth of a scope' with Oscium's iPad oscilloscope, which weighs only 0.085kg (3 ounces). The iPad oscilloscope attaches to the charging port of the iPad. And, it does not have a battery; it uses power from the tablet so there is no need for an outlet. With the small size and no need for charging, this sailor is agile enough to address all repair issues on the ship, regardless of where it is located.

—Bryan Lee, president, Oscium



high-end MSOs had the capability of serial triggering, but it is slowly trickling down to basic MSOs offering 70MHz and 100MHz bandwidths.

"Designers of embedded systems operating at 20MHz and lower are addressing several applications including the IoT and wearable devices," says Karl Chen, product manager, Rigol Technologies Inc. He adds, "This makes MSOs very affordable and provides great value, especially for designers working on a plethora of small- and mid-level embedded designs."

For instance, MSO1000Z series from Rigol and SDS2000 series from Siglent provide serial triggering functions at an affordable price. See Table for pricing and other details.

Interesting developments in portable, pocket-sized MSOs

USB based tools used with PCs or laptops. When engineers look for flexibility, portability and ease of measurement, universal serial bus (USB) oscilloscopes prove to be a practical option.

USB MSOs are mostly used by

engineers and technicians working in service, testing, equipment repair and manufacturing. Some MSOs are connected via USB, Ethernet or even RS232 and are widely used for a variety of applications starting from research and development (R&D) to embedded designs and educational purposes.

An MSO integrated with a protocol analyser enabled with USB 3.0 and Wi-Fi capabilities was recently introduced; a first of its kind in the PC based test instrument group.

USBee QX comes with a bandwidth of 600MHz and a combination of four analogue and 24 digital channels. One unique feature is that it can record 32 days of traces and display protocols as it has a buffer memory of 896 Msamples.

An added benefit is that this device includes all popular serial protocols that usually need to be paid for separately in the case of benchtop MSOs.

Another option for hackers, hobbyists, students and engineers is a tool nearly the size of a Snickers chocolate bar. This device from BitScope comes with 20MHz bandwidth, which is apt for small- and,

probably some, medium-level embedded applications. Completely user programmable, BitScope Micro also features two analogue channels and six digital channels. It also behaves as an 8-channel logic analyser, arbitrary waveform, a clock generator and spectrum analyser. In addition to Windows, Mac and Linux, it is also compatible with Raspberry Pi including its latest versions, Model A+ and Model B/B+.

Tool specifically for Apple phones and tablets. An answer to test-bench tools with mobile and touchscreen advantage is Oscium's iMSO-204L, a plug-and-play solution. Although specifically designed only for Apple devices, such solutions have become a viable alternative if engineers prefer to switch over to touch interface from traditional knobs.

These transform iPads or iPhones into a variety of test equipment tools like spectrum analysers, power meters, logic analysers and oscilloscopes. These are ultra-portable and small enough to fit in a pocket.

Bryan Lee, president, Oscium, says, "It is powered by the host device with less than 100mA of current (less than what it takes to power a small LED light), so there is very little impact to the battery life, eliminating the need for an external power source."

He adds, "The software is very easy-to-use. It can be downloaded from Apple App Store and viewed in demo mode without access to hardware. After hardware is connected, field engineers can monitor, store, and email results while on-the-go." He also claims that no other oscilloscope offers a solution quite like this unique system.

Quite recently, three software updates were made. As a result, users can now make advanced measurements between cursors, observe signal clips indicated by a colour change and can trigger in rolling mode.

Lee notes, "Not every user will be impacted by these changes but it demonstrates that we are continually investing in the tool and listening to our users." ●